Q1 Reverse a list

1 mark

Write a C program that takes an array as input and returns it reversed.

Your Code

```c
#include <stdio.h>
int main(){
    int n, i, a[100];
    scanf("%d",&n);
    for(i=0; i < n; i++)
        scanf("%d",&a[i]);
    for(i=n - 1; i >= 0; i--)
        printf("%d", a[i]);
    return 0;
}
```

Input

```
5
1 2 3 4 5
```

Output

```
54321
```

Submitted

Your Code

```c
#include <stdio.h>
int main(){
    int n,i,key, low, high, mid;
    int a[100];
    for(i=0; i< n;i++)
        scanf("%d",&key);
    low= 0;
    high = n-1;
    while(low <= high){
        mid = (low+high)/2;
        if(a[mid] < key){
            printf("Found");
            return 0;
        }
        else if(a[mid] < key)
            low = mid+ 1;
        else
            high = mid - 1;
    }
    printf("not Found");
    return 0;
```

Input

```
5
10 20 30 40 50
30
```

Output

```
not Found
```

Q3 Searching

1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```
#include <stdio.h>
int main(){
    int n,i,key, low, high, mid;
    int a[100];
    for(i=0; i< n;i++)
        scanf("%d",&key);
    low= 0;
    high = n-1;
    while(low <= high){
        mid = (low+high)/2;
        if(a[mid] < key){
            printf("Found");
            return 0;
        }
        else if(a[mid] < key)
            low = mid+ 1;
        else
            high = mid - 1;
    }
    printf("not Found");
    return 0;
```

Input

```
5
10 20 30 40 50
30
```

Output

```
not Found
```

Q4 **Searching** 1 mark

Write a C program for Count the Occurrences of an Element

Your Code

```
#include <stdio.h>
int main(){
    int n,i,x,count = 0;
int a[100];
scanf("%d",&n);
for(i= 0; i< n; i++)
    scanf("%d",&a[i]);
scanf("%d",&x);
for(i=0;i<n;i++){
    if(a[i]==x)
        count++;
}
printf("%d",count);
return 0;
}
```

Input

```
6
1 2 2 3 2 4
2
```

Output

```
3
```

Submitted

Q5 Array

Write a program for binary search

Your Code

```
#include <stdio.h>
int main(){
    int n,i,key,low,high,mid;
    int a[100];
    scanf("%d,&n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&key);
    low = 0;
        high = n - 1;
    while(low<=high){
        mid=(low+high)/2;
        if(a[mid]==key){
            printf("found");
            return 0;
        }
        if(a[mid]<key)
            low=mid+1;
```

Input

```
5
10 20 30 40 50
30
```

Output

```
not found
```

Q6 **Merging** 1 mark

Write a C program for Merge Two Arrays

Your Code

```c
#include <stdio.h>
int main()
{
    int n,m,i;
    int a[100],b[100];
    scanf("%d",&n);
    for(i = 0;i < n; i++)
        scanf("%d",&a[i]);
    scanf("%d",&m);
    for(i=0;i<m;i++)
        scanf("%d",&b[i]);
    for(i = 0;i<n;i++)
        printf("%d",a[i]);
    for(i = 0;i<m;i++)
        printf("%d",b[i]);
    return 0;
}
```

Input

```
3
1 2 3
3
4 5 6
```

Output

```
123456
```

Q7 Largest and Smallest element

1 mark

Write a C program for Find the Largest and Smallest Element in an Array

Your Code

```c
#include <stdio.h>
int main()
{
    int n,i,max,min,a[100];
    scanf("%d",&n);
    for(i = 0;i < n;i++)
        scanf("%d",&a[i]);
    max = min =a[0];
    for(i=1;i<n;i++)
    {
        if(a[i]>max) max=a[i];
        if(a[i]<min) min=a[i];
    }
    printf("largest=%d\n",max);
    printf("smallest=%d",min);
    return 0;
}
```

Input

```
5
10 20 5 30 15
```

Output

```
largest=30
smallest=5
```

Submitted

Q3 **Searching** 1 mark

Write a C program for Searching an Element in a Sorted Array

Your Code

```c
#include <stdio.h>
int main(){
    int n,i,key, low, high, mid;
    int a[100];
    for(i=0; i< n;i++)
        scanf("%d",&key);
    low= 0;
    high = n-1;
    while(low <= high){
        mid = (low+high)/2;
        if(a[mid] < key){
            printf("Found");
            return 0;
        }
        else if(a[mid] < key)
            low = mid+ 1;
        else
            high = mid - 1;
    }
    printf("not Found");
    return 0;
```

Input

```
5
10 20 30 40 50
30
```

Output

```
not Found
```

Q9 **Occurrence of an elemnet** 1 mark

Write a C program for Find the First and Last Occurrence of an Element

Your Code

```c
#include <stdio.h>
int main(){
    int n,i,x,first=-1,last=-1;
    int a[100];
    scanf("%d",&n);
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
    scanf("%d",&x);
    for(i=0;i<n;i++)
    {
        if(a[i]==x){
            if(first==-1)
                first=i;
            last=i;
        }
     }
    printf("%d%d",first,last);
    return 0;
```

Input

```
6
1 2 3 2 4 2
2
```

Output

```
15
```

Q10 ## Missing Element

1 mark

Write a C program for Find Missing Element in an Array

Your Code

```c
#include <stdio.h>
int main()
{
    int n,i,sum=0,total;
    scanf("%d",&n);
    int a[n];
    for(i=0;i<n;i++)
    {
        scanf("%d",&a[i]);
        sum=sum+a[i];
    }
   total=(n+1)*(n+2)/2;
    printf("%d",total-sum);
    return 0;
}
```

Input

```
4
1 2 4 5
```

Output

```
3
```

Submitted